

| M1    | M2     | Test set | $n$ | $\Delta_B$ | $\hat{P}_0$ | $\hat{b}_0$ |
|-------|--------|----------|-----|------------|-------------|-------------|
| TF19* | TF18*  | 2019     | 2K  | 4.3        | 0.19        | 23.7        |
| TF18  | TF16   | 2018     | 3K  | 4.2        | 0.09        | 29.4        |
| TF16  | Conv17 | 2017     | 3K  | 1.3        | 0.12        | 22.5        |
| TF16  | Conv14 | 2016     | 3K  | 7.6        | 0.10        | 27.6        |

Table 3: Relevant parameters from four MT evaluations. TF are Transformer-based (Ott et al., 2018; Edunov et al., 2018; Ng et al., 2019) and Conv are Convolutional models (Gehring et al., 2017) from FAIRSEQ. Test sets are from WMT shared tasks for En-De translation.  $\Delta_B$  is the reported difference in BLEU, whereas  $\hat{P}_0$  and  $\hat{b}_0$  are estimated. \* indicates ensembles.

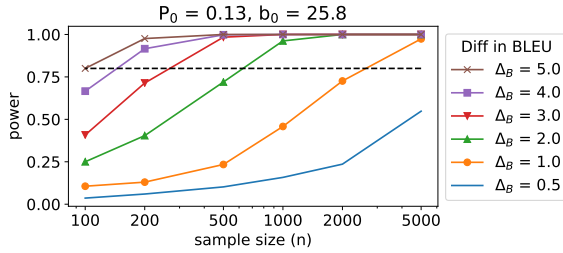


Figure 4: Power analysis for MT, showing how power increases with  $n$  and  $\Delta_B$ , using an average of fitted values for  $P_0$  and  $b_0$ . Based on this analysis, we expect that an experiment with a test set of 2000 sentences would have approximately 75% power to detect a difference of 1 BLEU point as significant. For additional plots, refer to Figure 17 in Appendix G.

While far from identical, the four comparisons, each representing different stages of model evolution, all produce similar estimates. Although these estimates are only based on a single language pair, the models and test sets are relatively diverse, and we expect that these estimates will generalize, though better estimates could be obtained by fitting this distribution to a new domain of interest.

Using these estimates, we can now characterize how much power test sets of different test set sizes ( $n$ ) would have for a range of possible differences in BLEU ( $\Delta_B$ ). Figure 4 shows this for  $P_0$  and  $b_0$  set to the average of the observed values.<sup>10</sup> Based on this estimate, we conclude that for typical MT test sets of around 2,000 examples, an improvement of 1 BLEU point can likely be detected with approximately 75% power. As shown in Figure 4 this power level increases dramatically with sample size and effect size.

This analysis has served, in part, to show how a simulation-based approach to power analysis can

<sup>10</sup>For a sensitivity analysis of how power varies under different assumptions for  $P_0$  and  $b_0$ , please see Figure 17 in Appendix G.

be adapted to virtually any task. Additional work is required to test how well these specific parameter estimates will generalize, but the same process can easily be adapted to new language pairs. More generally, there would be great value in the MT community curating larger held-out test sets, both to validate this analysis, and for better powered future comparison.

## 5 Likert-Scale Human Evaluations

Tasks such as natural language generation are difficult to evaluate using automated methods; as such, human evaluations are central to NLP. Past work has reported great variation in how human evaluations are done (van der Lee et al., 2019). Therefore, we begin with a meta-analysis of a subset of human evaluation experiments from EMNLP 2019, which we then use as the basis for claims about the power of human evaluations in NLP more generally.

### 5.1 Meta-analysis

To characterize the state of human evaluation in NLP, we identified papers from the main session of EMNLP 2019 that made use of human evaluations (details in Appendix H.2). To generalize across studies, we restrict our analysis to Likert-scale comparisons, which was the most commonly reported type of evaluation. We extracted all cases where a new model was being compared to the best-performing baseline on one more metrics (117 comparisons from 41 papers) and normalized all ratings to be on a 0-1 scale.

One takeaway from this meta-analysis is that the reported effect sizes (that is, difference between the novel model and the best-performing baseline) vary widely (s.d. = .12 on a [0, 1] scale). Number of items tested is more consistent: 69% used 100 or fewer, and only 18% used over 200. But, as similarly found by van der Lee et al. (2019), many key details were not reported in this sample of experiments. Most commonly missing was number of ratings per item (34% of all experiments), followed by total number of workers (28%). For 7% of experiments, we could not determine the number of items tested. 57% of experiments collected 3 annotations per item, which was also the modal number of unique annotators. Thus, it is often difficult to ascertain, for any particular experiment, the details of the experimental setting that are necessary to evaluate the validity of the results.

Because the number of items rated was the most

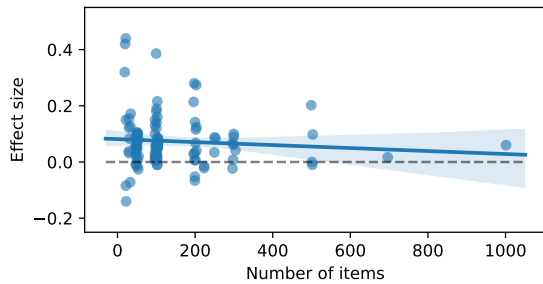


Figure 5: Scaled effect size vs. number of items from our EMNLP 2019 survey, showing higher variance in the smallest samples. There is a slight negative correlation, though it is not significant. As can be seen, most experiments are small ( $n \leq 100$ ).

commonly reported, we use that as our proxy for sample size. Figure 5 shows scaled mean difference between models as a function of number of items. As expected, we see greater variance in effects with smaller samples since, with smaller samples, we expect greater noise. We also observe a slight negative correlation between effect size and sample size. That is, as sample size gets larger (and, thus, as estimates get more precise), the estimated effect size gets smaller. This trend is sometimes used as an indication of publication bias (censoring of null and opposite-direction effects) since, in a sample with no publication bias, the effect size should be independent of the sample size (Begg and Mazumdar, 1994). However, in our case, this correlation is not significant (Kendall’s  $\tau = -.07$ ,  $p = .32$ ) and so it is difficult to draw strong conclusions.<sup>11</sup>

## 5.2 Power analysis for human Likert ratings

What kind of effect sizes can typical human evaluation experimental designs detect? As in previous sections, we can use simulations to explore how many annotators and/or instances should be used to have sufficient power.

Simulating human experiments is conceptually simple (e.g.,  $m$  raters each rate  $n$  generated sentence on overall quality), but for realistic simulations, we need to consider variation in items (some generated sentences are better than others), and variation by rater (some raters use higher ratings and/or respond to different aspects of quality), as well as the overall difference in quality between models. A simulation which treated all workers as identical would fail to capture this variation, and hence might overestimate power (Barr et al., 2013).

<sup>11</sup>We exclude from this analysis two large negative effects with  $N = 500$  which would exaggerate this correlation.

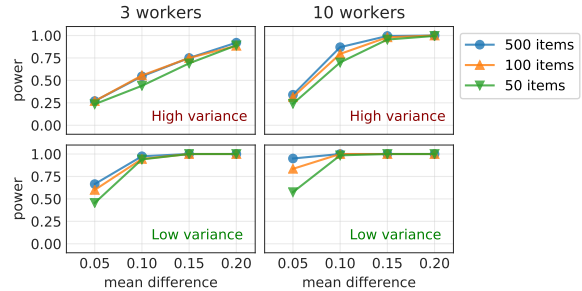


Figure 6: Using parameters estimated with mixed effects models from a high variance setting (top) and a low variance setting (bottom), the left panel shows simulated experiments with 3 workers annotating each item, the right panel shows an unusually high number of annotators per item (10 workers). Under typical assumptions, many common experimental settings (e.g., 3 workers and 100 items) are underpowered.

Unfortunately, details such as worker variance are rarely reported in published papers. To better characterize the typical variation in human evaluations, we rely on a convenience sample of several large datasets to estimate these parameters and use them in our simulations as a proxy for what we might observe in practice. Although focused on different tasks, all use a similar methodology, namely, getting many Likert-scale annotations per instance from many annotators and models (in some cases as many as 20 ratings per item).<sup>12</sup>

In order to extract estimates of these parameters for our simulations, we use hierarchical mixed-effects models, as used in psychology and other behavioral fields (Barr et al., 2013; Gelman and Hill, 2006). Such models incorporate variation in the quality of generated instances, annotator responses, and annotator sensitivity, and are recommended by van der Lee et al. (2019) for analyzing human evaluations. (We provide details in Appendix H.3 and include code for fitting such models as part of the online materials). Using this approach, we obtain an estimate of the relevant parameters from each of the large datasets. From these, we choose sets of parameters to be representative of experiments with high or low variance, with full results in Appendix H.3 (see Table 16 for parameter estimates).

As before, we then use these estimates to simulate data, assess significance on the simulated data (here using mixed effect regression), and compute power as a function of mean difference and sample

<sup>12</sup>We use publicly available or author-provided data from Hashimoto et al. (2019); Dathathri et al. (2020); Holtzman et al. (2020), and WMT19 (links in Appendix H.2).

size.<sup>13</sup> The resulting power estimates are shown in Figure 6, plotted in terms of effect size, sample size, and numbers of workers and items, for both the high and low variance scenarios. From this analysis, we highlight a few key takeaways:

- *Many human evaluation studies are likely underpowered:* Using the “high variance” parameters (which are typical of most of the datasets we used), the most common design at EMNLP 2019 (3 workers, 100 items) is underpowered unless the effect size is quite large (0.2 or higher on the [0, 1] scale).
- *Even with low variance, typical designs are underpowered to detect small effects:* Using our estimated parameters for the low variance setting, experiments will be underpowered to detect small effects (0.05 on the [0, 1] scale), unless an unusually large number of ratings per item are collected (10+ for 100 items).
- *Need for improved reporting:* Most human evaluations do not report enough detail to interpret the results. This could be drastically improved through basic power analyses, significance testing using mixed-effects models, and sharing of raw data.

Given our model estimates and simulations, we conclude that, in aggregate, many human evaluations are underpowered and would benefit from larger sample sizes, particularly by using more workers per item. Increased adoption of even approximate power calculations within the NLP community will promote thoughtful consideration of appropriate sample sizes and improve the reliability and replicability of results.

## 6 Overall Recommendations

- Power analyses should be done prior to evaluation when comparing against a baseline. If a comparison is likely to be underpowered, the pros and cons of running that evaluation should be carefully considered. Underpowered experiments do not provide convincing evidence of progress.
- For new datasets and shared tasks, the number of instances in the test will determine the

minimum detectable effect size, and should be chosen accordingly.

- For tasks which no longer have adequate power to detect typical improvements (e.g., MRPC and SST-2), authors should consider expanding the test set or retiring the task.
- To facilitate future power calculation and significance tests, model owners should release final fine-tuned model checkpoints. Alternatively, leaderboard owners may wish to make validation set predictions from all submitted models publicly available.
- For human evaluations, (anonymized) raw data should be shared, along with parameters and code to replicate the analysis, including proper significance testing. Prior to collecting human evaluation data, researchers should create an analysis plan and run power analyses to determine an appropriate sample size (likely requiring more workers and items than is currently typical in NLP).

## 7 Conclusion

Recent progress in NLP has been extraordinarily rapid, sometimes at the cost of experimental rigor. In this paper, we have presented evidence that underpowered experiments are widespread in NLP. For comparisons based on small samples, there is little reason to think that such an evaluation *could* reliably provide evidence of a significant improvement, and good reason to believe that improvements found to be significant will exaggerate or reverse the true effect. Going forward, a combination of larger test sets, simple power analyses, and wider sharing of code, data, and experimental details will help to build the foundation for a higher standard of experimental methodology in NLP.

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<sup>13</sup>These simulations require estimates for 7 parameters: the baseline, the effect size, variance by worker, variance by worker as a function of model, variance by item, variance by item as a function of model, and residual variance.